A Tale of Two Modalities for Video Captioning

Pankaj Joshi Chitwan Saharia Vishwajeet Singh
Digvijaysingh Gautam Ganesh Ramakrishnan Preethi Jyothi
Department of CSE, Indian Institute of Technology Bombay

Introduction

- *Investigating:* Influence of modality – audio and visual inputs – on the quality of textual captions generated from videos
- *Investigating:* Importance of pretraining and a modality-specific weighing scheme for video captioning.
- *Illustrating:* Importance of the audio modality in conjunction with video cues through quantitative and qualitative evaluations on the MSRVTT dataset.

Model Specifics

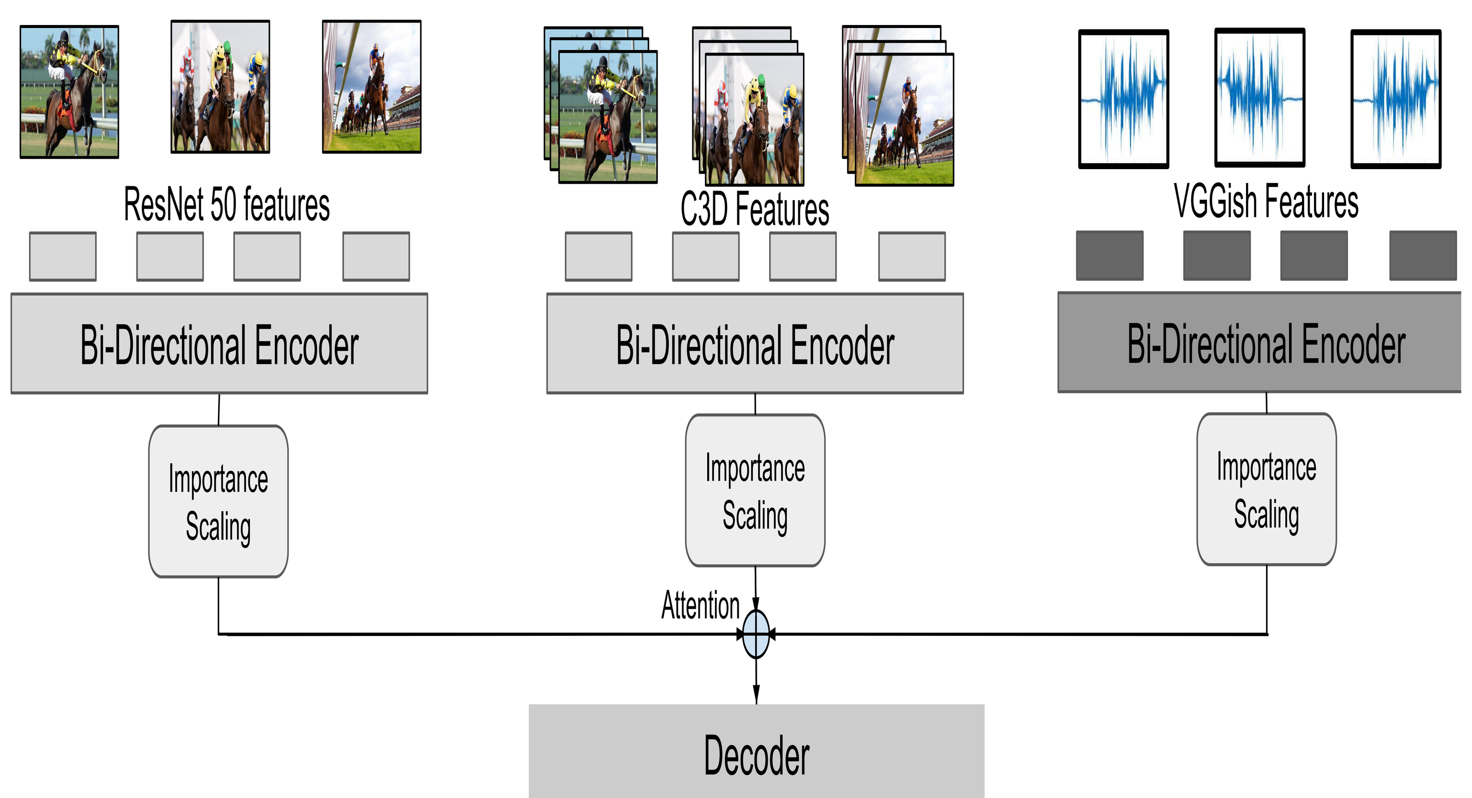

Fig. 1: Network Architecture

Architecture:

- Encoder-decoder with attention model, trained end-to-end using a cross-entropy loss function.
- BiLSTM-based encoders that use penultimate layer of Resnet50 and C3D as features for the visual modality and VGGish features for the audio modality.
- Attention + LSTM-based decoder to predict the next word in the caption.

Pretraining of encoders using an autoencoder-based objective function [4].

Importance score (i.e. a scaling factor $\in [0,1]$) learned for each modality at each time-step. These scores was used to scale the encoded representations for specific modalities.

Quantitative Analysis

System	BLEU-4	CIDEr	METEOR	ROGUE
A	33.1	25.6	23.7	56
A-PT	34.6	26.3	23.8	56.2
RCA	40.9	42.1	27.4	60.5
RC-PT	38.1	41.6	26.6	59.3
RC-PT+A	41	43.2	27.5	60.7
RC-PT+A-PT	**41.5**	**44.1**	**27.8**	**61.2**

Influence of pretraining on video captioning.

- **A** $\Rightarrow$ performance using only audio features and no video features. Pretraining the audio encoder (A-PT) gives clear improvements across all metrics.
- **RCA** $\Rightarrow$ model performance with using video features (Resnet) and (C3D) and VGGish audio features, without pretraining.
- **RC-PT + A** $\Rightarrow$ a pretrained video encoder and an audio encoder without pretraining. This gives consistent performance improvements compared to **RCA**.
- We see a consistent boost in performance by pretraining both video and audio encoders in **RC-PT + A**.

System	BLEU-4	CIDEr	METEOR	ROGUE
RC	38.1	41.6	26.6	59.3
RCA	41.5	44.1	27.8	61.2
RCA-IS	**41.8**	**44.3**	**27.8**	**61.4**
RCA-TOPIC	44.1	49.8	29	62.9
RCA-TOPIC-IS	44.3	50.4	28.9	62.7
SA-LSTM [6]	39.1	42.7	26.6	59.3
WEAKLY [3]	41.4	48.9	28.3	61.1
CIDENTRL [2]	40.5	51.7	28.4	61.4
HRL [5]	41.3	48.0	28.7	61.7
PICKNET [1]	41.3	44.1	27.7	59.8

Comparing pretrained video captioning models to existing models.

RCA-IS shows how importance scaling helps consistently improve performance when applied to different models. (**RCA-topic** shows oracle numbers if one could train an LDA-based topic model using ground-truth topics that are available with MSRVTT videos.) We show that our best system, **RCA-IS**, is comparable to state-of-the-art in prior work.

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Qualitative Analysis

System	F1	NDCG	MAP	MRR
A-PT	9.62	40.06	19.69	37.49
RC-PT	9.94	40.26	18.56	36.99
RC-PT + A	10.87	45.62	21.53	41.91
RC-PT + A-PT	10.65	43.98	21.65	41.06
RC-PT + A-PT + IS	**11.14**	**45.93**	**22.60**	**42.67**
RC-PT + A-PT + TOPIC	11.49	46.79	22.36	42.97

Pre-hoc analysis: For 130 test videos, three expert annotators identified keyphrases from the videos and partially ordered based on relevance to the video. The partial order is collapsed to count precision at each rank; an exact match of the keyphrase within a generated caption is given a score of 1 and 0, otherwise. Importance scaling tends to *significantly pull up important keyphrases*.

System	C	R	A	V
A-PT	**1.91**	0.74	0.89	0.82
RC-PT	1.86	0.96	0.86	1.05
RC-PT + A	1.85	1.03	0.94	1.14
RC-PT + A-PT	1.87	1.10	**1.01**	1.18
RC-PT + A-PT + IS	1.88	**1.12**	1.00	**1.18**
RC-PT + A-PT + TOPIC	1.89	1.13	1.02	1.22

Post-hoc analysis: A random subset of 100 test videos were evaluated by six annotators. Captions from the six systems above for these 100 videos were scored by the annotators using a 3-point scale for four metrics: Coherence(C), Relevance(R), Audio Evidence(A), Visual Evidence(V). The influence of importance scaling and pretraining are both highlighted in this post-hoc analysis.

References

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